

Customer Financial Behaviour and Risk Analysis

What I found in Morgan Stanley's customer transaction data, and what I'd recommend the bank do about it.

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The short version

I looked at 800 transactions from 189 customer accounts, covering about 18 months (January 2023 to June 2024). The bank wanted to understand how customers use their accounts, find any that look risky, and get some practical advice. I cleaned the data, grouped the accounts by activity and balance, built profiles of the customer types, and checked every account for warning signs. Out of 189 accounts, **seven stood out** — they set off three of my four risk flags. That small watchlist is the main takeaway.

One honest point runs through the whole report: in this dataset, the numbers — how much people transact, their balances, their risk scores — don't move together the way they normally would in real banking data. So a few patterns that look meaningful turn out to be side effects of the data's structure, not real behaviour. I've flagged those, because telling the difference is the whole job.

About the data

Each row is a single transaction, with the account, date, type (deposit, withdrawal, payment, transfer), amount, balance, and a risk score. In total there are 800 transactions across 189 accounts (about 4 each) and 180 customers. One thing I kept in mind: 2024 only runs to June, so it's a half year — I didn't compare the 2023 and 2024 totals as equals.

Cleaning the data

There were no missing values and no duplicate rows. Four judgement calls mattered:

- **The transaction ID was broken** — only 197 of 800 were unique, and the same ID sat on completely different transactions. So the ID was broken, not the rows. I kept every row and used AccountID as the key instead; deleting the "duplicates" would have thrown away about 600 real transactions.
- **22 amounts were negative**, some on deposits — but a negative deposit makes no sense, so I treated the sign as noise and took the absolute value. Direction is already captured by the transaction type.
- **The risk score ran from -0.30 to 1.25**, outside the usual 0-to-1 range, but it spilled past both ends evenly — the sign of a standardised score, not a broken percentage. I left it as-is; forcing it into 0-to-1 would have flattened the highest- and lowest-risk customers into the same value.
- **21 balances were negative**. These are real overdrafts, exactly the accounts a bank wants to find, so I kept them.

Worth noting: I corrected the negative amounts but kept the negative balances — I judged each by what it could actually mean, not by a blanket rule.

How customers move money

The data gives the transaction type but doesn't label money-in (credit) versus money-out (debit), so I mapped it myself: a deposit is money in; withdrawals, payments, and transfers are money out. The transfer one is an assumption (transfers usually go out), and I noted it can be flipped.

When I plotted money-in against money-out over the 18 months, money-out was higher in every month. That looks like customers steadily losing money, but it isn't real: three of my four types are "money out" and only one is "money in", so money-out draws on about three times as many transactions. It's a side effect of the labelling, not behaviour.

I also flagged **dormant** accounts. The literal "any two-month gap" rule flags 88% of accounts here just because transactions are sparse, so I used the way banks mean it — no activity in the last two months of the data — which flagged 121 of 189. I'd still read that cautiously, since quiet stretches are normal with so few transactions per account.

Customer profiles

For activity level I used fixed cut-offs (my rubric: **Low = 1–3 transactions, Medium = 4–6, High = 7+**), which gave **86 Low, 72 Medium, and 31 High**. The groups are uneven on purpose — only about 16% of accounts are highly active, and equal thirds would have hidden that. For balance I did the opposite and split into equal thirds by rank, because balance is skewed and a fixed cut-off based on the average would be unreliable.

Then the three profiles the bank asked for:

- **High-net-inflow accounts** (top 25% by money-in minus money-out). Because of the labelling side effect, this cut-off is actually negative, so it really means "the least-negative quarter" — though the very top accounts are genuinely positive.
- **High-frequency, low-balance accounts** — only 5 fit this. If the two traits were unrelated I'd expect about 10 by chance, so 5 is fewer than chance: not a real cluster, just random overlap.
- **Negative or near-zero balance accounts** — 22 accounts that dropped below 1,000 at some point. I looked at individual transactions, not averages, because most of these have a healthy average balance but still dipped into the red at least once, and that dip is what matters for risk.

Finding the risky accounts

This is the core of the report. **21 accounts overdrew** (went into the red) at least once — each exactly once, so the overdraft itself is the signal rather than how often.

For **large withdrawals** I used the top 10% of withdrawal amounts (about 89,000+). I considered defining "large" relative to each account's balance, but balance and amount are basically unrelated here (correlation -0.005), so that would just react to noise. Most were one-offs, but one account made two — the only repeated pattern.

For **balance swings** I used the coefficient of variation (the size of the swings compared to the account's typical balance, so big and small accounts compare fairly). I checked that the

most "volatile" accounts have healthy averages, so the swings are real and not a division artifact.

I also ran a standard outlier test (IQR) on amounts and net inflow and found **zero outliers**. That's a real result, not a mistake — outliers are about values that sit apart from the pack, not just big values, and this data is spread smoothly with nothing detached.

Finally I combined four signals — overdraft, large withdrawal, high balance swings, and high risk score — into a single count per account. Most accounts (96) had none, and the counts fall away steadily to just **seven accounts with three flags and none with all four**. Those seven are my priority watchlist — the ones I'd hand to the bank first. One caveat: because these signals are unrelated in this data, an account can collect several flags partly by chance. In real banking data they'd usually go together, and the same method would then point straight at the customers who need help.

The hypothesis test

The bank asked whether accounts that transact a lot hold higher balances than those that transact a little. I split the accounts at the median (4 transactions): the high-volume group (71 accounts) averaged about 74,600, and the low-volume group (118 accounts) about 70,400 — a gap of roughly 4,300, or about 6%. I ran a one-sided t-test (one-sided because the question asks about "higher") and it came out just past the usual cut-off for significant ($p \approx 0.04$).

So technically high-volume accounts do tend to have slightly higher balances — but it's weak: the result is right on the borderline, the difference is small (~6%), and it goes against my earlier finding that balance and activity are basically unrelated here. My honest read is weak, borderline evidence of a slight tendency, not something I'd build a decision on.

What I'd recommend

1. **Watch the seven flagged accounts** — give the watchlist a manual review.
2. **Use balance swings as an early warning** — big swings are a simple sign something may be changing.
3. **Reach out to overdraft accounts** — a friendly check-in before it happens again.
4. **Don't over-read the volume-and-balance link** — the connection is too weak to design strategy around.

More broadly, this data behaves like generated data, where the measures don't move together the way real customer data does. On real data these same methods would likely show stronger, clearer patterns.

My own criteria, in one place

- **Dormant account** = no activity in the final 60 days (60 days standing in for "two months"); chosen over the literal "any two-month gap", which flagged 88% of accounts.
- **Large withdrawal** = top 10% of withdrawal amounts (~89,000+); an absolute cut-off, since balance and amount are unrelated here.

- **Activity tiers** = Low (1–3), Medium (4–6), High (7+).
- **Balance tiers** = equal thirds by rank.
- **Near-zero / negative balance** = balance under 1,000 at any point.
- **Risk watchlist** = 3 or more of the 4 risk flags.